

Effect of Agnihotra on Life Satisfaction

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Abstract— This study intends to investigate whether the traditional Vedic fire ritual, Agnihotra, has any potential impact on an individual's life satisfaction. Agnihotra, is an ancient fire ritual originating from Vedic traditions in India and is practiced across several countries across the world. The current body of scientific research mostly ignores the effect of Agnihotra in raising life satisfaction. A considerable study gap exists when it comes to precisely evaluating the effect of Agnihotra on life satisfaction. The study primarily aims to explore how Agnihotra might improve life satisfaction. A questionnaire-based survey was conducted amongst Agnihotra practitioners and non-practitioners to investigate the effect of Agnihotra on life. The 5-item scale of Diener et al. (1985) is used to measure the judgments of one's life satisfaction with items of fulfilled, superb, content, successful, and fulgent. Participants indicate how much they agree or disagree with each of the 5 items using a 7-point scale that ranges from '7' meaning 'strongly agree' to '1' meaning 'strongly disagree'.

The study reports the effect of demographic variables such as gender, age, marital status, no. of family members, financial status on life satisfaction scores both on Agnihotra practitioners and non practitioners. Moreover, the results of the study indicated that Agnihotra practitioners have higher level of life satisfaction than non-practitioners amongst similar demographic variable categories.

Introduction

Agnihotra, an ancient Vedic fire ritual that has been carried out in India. A scientifically proven simple process of burning fire using dried cow-dung cakes and offering uncooked, cow ghee smeared rice grains, to the fire at the precise time of sunrise and sunset, with the chanting of two simple mantras in a pyramid-shaped copper vessel. Agnihotra is a tradition that has its roots in Vedic scriptures from the Rigveda, Yajurveda, Atharvaveda and even subtle mentions across the Buddhist literatures, Bible and Koran. The transformative effects of Agnihotra have been documented in numerous [scientific studies](#), personal experiences and testimonials around the world. It is a healing tool which anyone can utilize regardless of race, creed or religion. The whole process takes not more than five to ten minutes during morning and evening. Agnihotra is highly revered on a cultural and spiritual level, and its usefulness in resolving environmental and health issues is still being debated and researched.

Life satisfaction is a fundamental concept in psychology and social sciences, sometimes known as subjective well-being or happiness. It reflects an overall assessment of an individual of their life, considering both their cognitive assessments and emotional experiences. It goes beyond objective indicators like income, education, and health to capture the individual's perception of their well-being. Researchers attempt to understand the big picture of life satisfaction for several reasons, which include psychological well-being, health and longevity, social and economic outcomes, policy and governance, and positive psychology.

Comparing life satisfaction across cultures and countries can provide important insights into the differences in cultural values, norms, and priorities. It aids in the understanding of how socioeconomic and cultural issues impact well-being by researchers and policymakers.

I. LITERATURE REVIEW

Agnihotra resuscitated from the ashrams of Shivpuri, Akkalkot, Maharashtra, India is performed across borders by practitioners as a process to heal the atmosphere and this helps them to gain more mindfulness, develop an attitude of servitude, which helps them purify themselves spiritually and advance their spiritual growth and inner tranquillity. Agnihotra is highly revered on a cultural and spiritual level, yet its usefulness in resolving environmental and health issues is still being debated and researched.

A. Agnihotra: An Ancient Vedic Practice

Agnihotra origins may be traced to the Vedas, the earliest holy books known to mankind, which are said to have been written between 1500 and 1200 BC. The Yajur Veda, one of the four Vedas, contains a particular reference to Agnihotra in a portion known as the "Shukla Yajur Veda" and the "Krishna Yajur Veda". The Agnihotra ritual is described in detail in these scriptures.

B. Explanation of Agnihotra Ritual

The Sanskrit words "Agni" (fire) and "Hotra" (offering or sacrifice) are combined to get the term "Agnihotra". As a result, Agnihotra might be interpreted as a "fire offering" or "fire sacrifice". Agnihotra was historically one of the everyday rites carried out by Vedic priests and common people in ancient India. Agnihotra is a ritual that has a strong connection to the daily biorhythmic cycle of sunrise and sunset. It is performed at the precise local biological time of sunrise (Pratah Agnihotra) and sunset (Sayam Agnihotra). These timings are essential because they are the Transitional time of the phases of change from night to day and from day to night. At sunrise, there is release of bio-energy which helps cleanse both the surroundings and mentality. At sunset, one can harness the abundant energy through the agency of fire. This is believed to provide harmony and balance to all of nature including the performer. Agnihotra is a very precise rite which also has a profound spiritual significance. Its procedure is as follows:

- **Preparation:** The cow dung cakes are arranged in inverted pyramid shaped copper vessel and unbroken raw rice grains soaked in cow's (or its progeny) ghee (clarified butter) are made in preparation.

- **Timing:** Agnihotra must be started precisely at the time of local sunrise and local sunset, as was previously specified.
- **Lighting the Fire:** Fire is initiated by igniting the cow-dung cakes smeared with the cow's ghee within the pyramid. After starting the fire, the practitioner adds ghee to get the fire to a full blaze until the exact time of Sunrise / Sunset as applicable.
- **Offerings & Mantras:** At this precise time there are two offerings / oblations of uncooked unbroken rice grains smeared in cow's ghee; one after recitation of each Mantra (Manas – means Mind and 'tra' means instrument – Mantra essentially functions as a 'Tool of the Mind'); are offered into the fire as oblations during the rite. Exactly at sunrise/sunset time, accompanied by the chanting of sunrise/sunset Agnihotra Mantras – "Suryaya/Agnaye Svaha" offer the first part after saying Svaha and complete the Mantra by saying "Agnaye Idam Na Mama". Similarly, uttering the second line of the Mantra offer other part of the rice grains to the fire after saying "Prajapataye Svaha" and complete the Mantra uttering "Prajapataye Idam Na Mama". Concentrate on the fire till the offerings are fully burnt.
- **Silent Meditation:** After the oblations and mantras are finished, a period of silent meditation or prayer is observed, with the focus being on the sense of mindfulness and gratitude which leads to a spiritual connection.

C. Life Satisfaction

Life satisfaction is a complex and multifaceted concept that encompasses an individual's feelings, attitudes, and perceptions about life as a whole and is often used as a key indicator of an individual's well-being and quality of life. A person's entire level of mental and emotional health is referred to as their emotional well-being. This includes their capacity to manage stress, feel happy, be resilient in the face of difficulties, and cultivate healthy relationships.

Diener et al. (1985) [1] developed "The Satisfaction with Life Scale (SWLS)" as a measure of the judgmental component of subjective well-being, which was subsequently validated by independent studies. In comparison to many other measures of life satisfaction, the SWLS has been demonstrated to be a valid and reliable indicator of life satisfaction that is appropriate for use with a wide variety of age groups and applications.

II. METHODOLOGY

A. Define Research Objectives and Hypotheses:

This study intends to explore and determine the specific aspects of life satisfaction concerning Agnihotra.

Additionally, this study tests hypotheses that outline the expected findings based on the existing literature. The SWLS scores for each respondent are dependent variable, and whether they are Agnihotra practitioners or not will be the independent variable (i.e., a categorical variable with two levels: practitioners and non-practitioners).

Descriptive Statistics:

Mean, median, standard deviation for life satisfaction scores for both Agnihotra practitioners and non-practitioners are calculated. This gives you a preliminary sense of the differences in life satisfaction.

Hypothesis Testing (Statistical Tests):

Choose the appropriate hypothesis test compare the mean life satisfaction scores between the two groups. If SWLS scores in both groups are approximately normally distributed and variances of two groups are equal, **t-test** can be used. If not, **Mann-Whitney U test (Wilcoxon rank-sum test)** can be used. Normal distribution criteria can be checked using visual methods (histograms or Q-Q plots) or statistical tests like the Shapiro-Wilk test. Similarity in variances can be tested using Levene's Test for Equality of Variance.

t-test: Null Hypothesis (H_0): There is no significant difference in life satisfaction between Agnihotra practitioners and non-practitioners. Alternative Hypothesis (H_1): There is a significant difference in life satisfaction between Agnihotra practitioners and non-practitioners. On performing t-test, p-value is obtained and if found to be less than (0.05), we can reject the null hypothesis and conclude that there is a significant difference between the two groups.

Mann-Whitney U test (Wilcoxon rank-sum test): Null Hypothesis (H_0): The distributions of life satisfaction are the same between Agnihotra practitioners and non-practitioners. Alternative Hypothesis (H_1): The distributions of life satisfaction differ between Agnihotra practitioners and non-practitioners.

Co-Variate Analysis: Analysis of Covariance (ANCOVA) is done to test whether Agnihotra practice influences life satisfaction while controlling covariates like gender, age, marital status, number of family members, and financial status.

- **Significant Difference:** If the p-value is below 0.05 (or your chosen significance level), and the effect size is moderate or large, Agnihotra practice has an impact on life satisfaction.
- **No Significant Difference:** if the p-value is greater than 0.05, then we fail to reject the null hypothesis, meaning there is any significant difference between the life satisfaction of Agnihotra practitioners and non-practitioners.

B. Questionnaire Design & Pre-Testing:

This study develops a questionnaire that seeks to collect both classification and interval-scaled statements related to Agnihotra and life satisfaction. The study has ensured that questionnaire items are clear, unbiased, and easy to

understand. The study has included demographic questions for the classification of respondents based on gender, age, marital status, number of members in the family, financial status, performing Agnihotra, and years since performing Agnihotra to characterize the sample of Hindus. The study used a 7-point Likert scale (ranging from 1 to 7) for interval-scaled 5 statements to measure respondents' perceptions and attitudes. Although the scoring was kept continuous such that the scoring sums up scores on each item of the interval scale with cut-off benchmarks used as extremely satisfied for 31 – 35, satisfied for 26 – 30, slightly satisfied for 21 – 25, neutral for 20, slightly dissatisfied for 15 – 19, dissatisfied for 10 – 14, and extremely dissatisfied for 5 – 9. Respondents were also told to give their overall opinion on the performance of Agnihotra that influences their life regarding the 5 statements using the interval scale. These five statements of the SWLS were “In most ways, my life is close to my ideal” coded as “Fulfilled”, “The conditions of my life are excellent” coded as “Superb”, “I am satisfied with my life” coded as “Content”, “So far I have gotten the important things I want in life” coded as “Successful”, and “If I could live my life over, I would change almost nothing” coded as “Fulgent”.

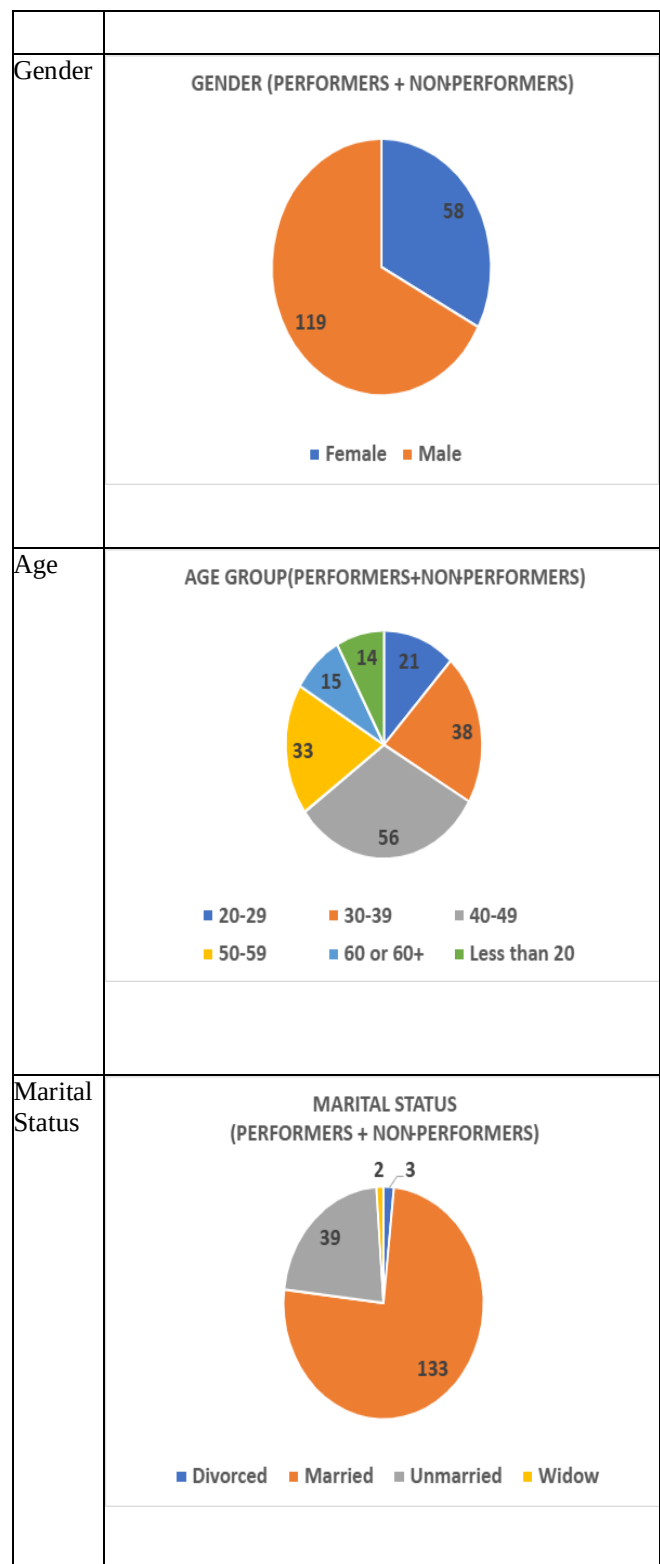
The questionnaire was made to undergo a pilot test with a small group of respondents, which happened to be 40 individuals, to identify any issues with question-wording, comprehension, or response options for making appropriate and duly required revisions.

C. Data Collection & Sampling Technique:

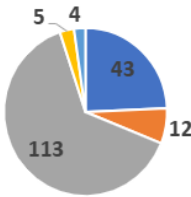
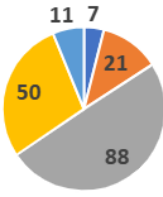
The data for this study was collected using an online filling of Google Forms with assurance to respondents that their responses would be kept anonymous and confidential while publishing the results in journals and conferences of repute. The target demographics were specified to be respondents who have been practitioners and non-practitioners of Agnihotra in Pune, Maharashtra. The study used the convenience sampling method making sure that the sample is varied and representative.

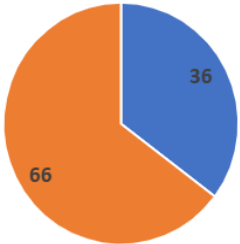
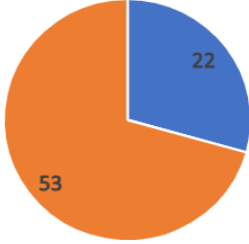
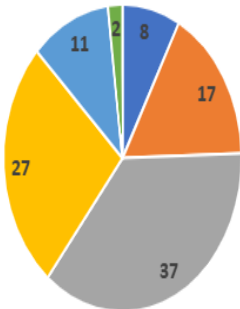
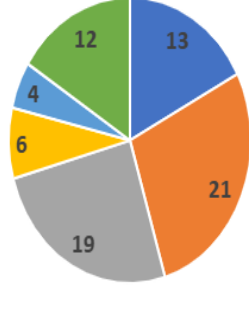
D. Data Analysis:

The usable data were to be analyzed using **MS Excel** to report summary statistics like means, frequencies, and percentages to analyze descriptive data. Based on a sample of **205-177** respondents, Table 1 provides demographic profile of respondents in terms of gender, age, marital status, number of family members, and financial status for both Agnihotra practitioners and non-practitioners. For Agnihotra practitioners, the distribution of the number of years individuals have been performing Agnihotra was also considered.



	Total (Performers + Nonperformers)
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Financial Strata	FINANCIAL STRATA (PERFORMERS + NON-PERFORMERS)  ■ Higher Middle Class ■ Lower Middle Class ■ Middle Class ■ Poor ■ Rich Class	NO. OF MEMBERS IN FAMILY (PERFORMERS + NON-PERFORMERS)  ■ 10 or 10+ Members ■ 1-2 Members ■ 3-4 Members ■ 5-6 Members ■ 7-9 Members
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	Performer	Non-Performer
Gender	GENDER (PERFORMERS)  ■ Female ■ Male	GENDER (NON-PERFORMERS)  ■ Female ■ Male
Age	AGE GROUP (PERFORMERS)  ■ 20-29 ■ 30-39 ■ 40-49 ■ 50-59 ■ 60+ ■ Less than 20	AGE GROUP (NON-PERFORMERS)  ■ 20-29 ■ 30-39 ■ 40-49 ■ 50-59 ■ 60+ ■ Less than 20

Marital Status	MARITAL STATUS (PERFORMERS) <table><tr><td>Divorced</td><td>1</td></tr><tr><td>Married</td><td>88</td></tr><tr><td>Unmarried</td><td>12</td></tr><tr><td>Widow</td><td>1</td></tr></table>	Divorced	1	Married	88	Unmarried	12	Widow	1	MARITAL STATUS (NON-PERFORMERS) <table><tr><td>Divorced</td><td>2</td></tr><tr><td>Married</td><td>45</td></tr><tr><td>Unmarried</td><td>27</td></tr><tr><td>Widow</td><td>1</td></tr></table>	Divorced	2	Married	45	Unmarried	27	Widow	1				
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Financ-- ial Strata	FINANCIAL STRATA (PERFORMERS) <table><tr><td>Higher Middle Class</td><td>22</td></tr><tr><td>Lower Middle Class</td><td>5</td></tr><tr><td>Middle Class</td><td>71</td></tr><tr><td>Poor</td><td>2</td></tr><tr><td>Rich Class</td><td>2</td></tr></table>	Higher Middle Class	22	Lower Middle Class	5	Middle Class	71	Poor	2	Rich Class	2	FINANCIAL STRATA (NON-PERFORMERS) <table><tr><td>Higher Middle Class</td><td>21</td></tr><tr><td>Lower Middle Class</td><td>7</td></tr><tr><td>Middle Class</td><td>42</td></tr><tr><td>Poor</td><td>2</td></tr><tr><td>Rich Class</td><td>3</td></tr></table>	Higher Middle Class	21	Lower Middle Class	7	Middle Class	42	Poor	2	Rich Class	3
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No.Of Family members	NO. OF MEMBERS IN FAMILY (PERFORMERS) <table><tr><td>10 or 10+ Members</td><td>2</td></tr><tr><td>1-2 Members</td><td>3</td></tr><tr><td>3-4 Members</td><td>59</td></tr><tr><td>5-6 Members</td><td>29</td></tr><tr><td>7-9 Members</td><td>9</td></tr></table>	10 or 10+ Members	2	1-2 Members	3	3-4 Members	59	5-6 Members	29	7-9 Members	9	NO. OF MEMBERS IN FAMILY (NONPERFORMERS) <table><tr><td>10 or 10+ Members</td><td>3</td></tr><tr><td>1-2 Members</td><td>5</td></tr><tr><td>3-4 Members</td><td>39</td></tr><tr><td>5-6 Members</td><td>25</td></tr><tr><td>7-9 Members</td><td>3</td></tr></table>	10 or 10+ Members	3	1-2 Members	5	3-4 Members	39	5-6 Members	25	7-9 Members	3
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Table 1 Descriptive Statistics

Parameter	N	Mean	Std. Error	Std. Devia	Variance
Age	177	41.39548	1.000358	13.3089	177.1268
Family Me	177	4.751412	0.193389	2.572872	6.619671
Fulfilled	177	5.141243	0.106821	1.421165	2.01971
Superb	177	5.237288	0.108693	1.446064	2.091102
Content	177	5.474576	0.106017	1.410463	1.989407
Successful	177	5.293785	0.112565	1.49758	2.242745
Fulgent	177	4.39548	0.136486	1.815834	3.297252

Table 2 Descriptive Statistics Total data

Table 2 provides descriptive statistics describing demographic variables and scaled variables named “Fulfilled”, “Superb”, “Content”, “Successful”, and “Fulgent” of the SWLS. There are 119 male respondents and the remaining are female respondents in Table 2. The mean age of the respondents in Table 2 is 41.39 years, indicating their average age. The standard deviation value of 13.30 years gives a measure of the amount of variation or dispersion in the ages. The standard error is also provided, which is useful for estimating the precision of the mean. The variance value of 177.126 is another measure of the spread of the age values. There are 39 unmarried respondents and 133 are married, 2 widow and 3 divorced respondents in Table 2. The average number of family members in the sample respondents is 4.75 with the value for standard deviation as 2.57 and the value for variance as 6.61. There are 5 respondents of poor, 12 respondents of lower middle class, 113 of them from the middle class, 43 of them from the higher middle class and the remaining are the rich class respondents in Table 2. The mean for “Fulfilled” is 5.14 with the values for standard deviation as 1.42 and variance as 2.01. The mean for “Superb” is 5.23 with the values for standard deviation as 1.44 and variance as 2.091. The mean for “Content” is 5.47 with the values for standard deviation as 1.410 and variance as 1.989. The mean for “Successful” is 5.29 with the values for standard deviation as 1.497 and variance as 2.242. The mean for “Fulgent” is 4.39 with the values for standard deviation as 1.815 and variance as 3.297.

	N	Mean	Std. Error	Std. Devia	Variance
Age	102	45.4706	1.1767	11.8845	141.2417
Family Me	102	4.7647	0.2243	2.2654	5.1322
Fulfilled	102	5.6078	0.1408	1.4223	2.0229
Superb	102	5.6373	0.1493	1.5077	2.2731
Content	102	5.9216	0.1259	1.2718	1.6176
Successful	102	5.6961	0.1506	1.5207	2.3127
Fulgent	102	4.7941	0.1861	1.8792	3.5315

Table 3 Descriptive Statistics Performers

Table 3 provides descriptive statistics describing demographic variables and scaled variables named “Fulfilled”, “Superb”, “Content”, “Successful”, and “Fulgent” of the SWLS. There are 66 male respondents and the remaining are female respondents in Table 3. The mean age of the respondents in Table 3 is 45.47 years, indicating their average age. The standard deviation value of 11.88 years gives a measure of the amount of variation or dispersion in the ages. The standard error is also provided, which is useful for estimating the precision of the mean. The variance value

of 141.24 is another measure of the spread of the age values. There are 12 unmarried respondents and 88 are married, 1 widow and 1 divorced respondent in Table 3. The average number of family members in the sample respondents is 4.76 with the value for standard deviation as 2.26 and the value for variance as 5.1322. There are 2 respondents of poor, 5 respondents of lower middle class, 71 of them from the middle class, 22 of them from the higher middle class and the remaining are the rich class respondents in Table 3. The mean for “Fulfilled” is 5.60 with the values for standard deviation as 1.420 and variance as 2.022. The mean for “Superb” is 5.63 with the values for standard deviation as 1.507 and variance as 2.273. The mean for “Content” is 5.921 with the values for standard deviation as 1.271 and variance as 1.617. The mean for “Successful” is 5.696 with the values for standard deviation as 1.520 and variance as 2.312. The mean for “Fulgent” is 4.794 with the values for standard deviation as 1.879 and variance as 3.531.

	N	Mean	Std. Error	Std. Devia	Variance
Age	75	35.8533	1.5252	13.209	174.4782
Family Me	75	4.7333	0.3413	2.9561	8.7387
Fulfilled	75	4.5067	0.1334	1.1552	1.3344
Superb	75	4.6933	0.1342	1.1622	1.3506
Content	75	4.8667	0.1581	1.3689	1.8739
Successful	75	4.7467	0.1484	1.285	1.6512
Fulgent	75	3.8533	0.1828	1.5828	2.5052

Table 4 Descriptive Statistics Non-Perform

Table 4 provides descriptive statistics describing demographic variables and scaled variables named “Fulfilled”, “Superb”, “Content”, “Successful”, and “Fulgent” of the SWLS. There are 53 male respondents and the remaining are female respondents in Table 4. The mean age of the respondents in Table 4 is 35.853 years, indicating their average age. The standard deviation value of 13.20 years gives a measure of the amount of variation or dispersion in the ages. The standard error is also provided, which is useful for estimating the precision of the mean. The variance value of 174.478 is another measure of the spread of the age values. There are 27 unmarried respondents and 45 are married, 1 widow and 2 divorced respondents in Table 4. The average number of family members in the sample respondents is 4.733 with the value for standard deviation as 2.95 and the value for variance as 8.73. There are 3 respondents of poor, 7 respondents of lower middle class, 42 of them from the middle class, 21 of them from the higher middle class and the remaining are the rich class respondents in Table 4. The mean for “Fulfilled” is 4.50 with the values for standard deviation as 1.155 and variance as 1.334. The mean for “Superb” is 4.69 with the values for standard deviation as 1.162 and variance as 1.350. The mean for “Content” is 4.86 with the values for standard deviation as 1.368 and variance as 1.873. The mean for “Successful” is 4.74 with the values for standard deviation as 1.285 and variance as 1.651. The mean for “Fulgent” is 3.85 with the values for standard deviation as 1.582 and variance as 2.505.

Comparing the mean values for SWLS parameters for performers and non-performers (table 3 and 4), it is evident that the life satisfaction level of performers is higher than that of non-performers.

SWLS Variab	U Value (Perfor	U Value (Non	Z - Value	P - Value	CLES
Fulfilled	956.5	2923.5	4.7779	1.77E-06	0.75
Superb	1082.5	2877.5	4.2764	1.9E-05	0.73
Content	917	3043	5.1079	3.26E-07	0.77
Successful	7813	1917	4.2242	2.4E-06	0.72
Fulgent	1239.5	2720.5	3.4928	0.000478	0.69

Table 5

Table 5 Provides a summary about Hypothesis Testing done for the different values of SWLS variables i.e. Fulfilled, Superb, Content, Successful and Fulgent for Performers and Non – Performers of Agnihotra. At Start, A Check for the Normality of Distribution for all the variables have been done to decide which test to be done for the Hypothesis testing and it is found that any of the distribution of a variable from SWLS is not normally distributed. As a Result, “Mann – Whiteny Test” is has been done to analyze distribution of values for the SWLS variables. For Every Variable the U Values (sum of ranks for both the tables performers and Non-performers), Z – Value, P – Value and Common language effect size is calculated. The P –Value for ‘Fulfilled’, ‘Superb’, ‘Content’, ‘Successful’, ‘Fulgent’ are 1.77E-06, 1.9E-05, 3.26E-07, 2.4E-06, 0.000478 which are significantly less than alpha, which indicates the randomly selected value of Performers' population is considered to be not equal to the randomly selected value of ‘Non-performers' population. In other words, the difference between the randomly selected value of Performers and the Non-performers populations is big enough to be statistically significant. The common language Effect Size (CLES) for different Variables are 0.75, 0.73, 0.77, 0.72, 0.69 which indicates the probability that a random value from the given SWLS variables for Agnihotra performers is greater than a random value for Agnihotra non performers.

CONCLUSION

This research aims to verify the effect of Agnihotra in fostering life satisfaction. This study has contributed to providing empirical evidence on the relationship between Agnihotra practice and life satisfaction. Findings from this study may have practical implications for individuals seeking alternative approaches to enhance life satisfaction. It may also guide practitioners, therapists, and spiritual leaders in incorporating Agnihotra into wellness practices. The dataset

reflects a consistent and positively rated perception of life satisfaction among the respondents.

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